1. Understand the relationship between the context of the problem and representation of the mathematical solution (Point 9)

Analysis: This is the core of the task. Students must understand what the "percentage" representation means in this specific real-world context: the ratio between "TV-owning households" and "total households." They must use this understanding to link a mathematical operation (division) to the real-world entity they are searching for (the total number of Swiss households).

2. Interpret a mathematical result back into the real world context (Point 1)

Analysis: The question does not ask for an abstract number, but an estimate of "total households in Switzerland." After performing the calculation ($2.8 \text{ million} \div 0.858 \approx 3.26 \text{ million}$), the student must interpret this result back into the context of the provided options to select the most realistic estimate (C. 3.3 million).

3. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (Point 2)

Analysis: When selecting an answer, students need to gauge the reasonableness of their result. For instance, since only 85.8% of households own a TV, the total number of households **must** be greater than 2.8 million. Using this contextual logic, a student can immediately rule out Option A and verify that 3.26 million is a sensible conclusion.

4. Interpret mathematical outcomes in a variety of formats in relation to a situation or use (Point 5)

Analysis: The student must process and navigate different data formats: text descriptions, tabular data, and percentages. They must evaluate which column in the table is relevant to the specific goal (total households) versus irrelevant data (cable subscription rates) to reach the correct outcome.

5. Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions (Point 8)

Analysis: The prompt asks for the "closest estimate." This reflects an understanding of the limits of mathematical solutions in statistics. The student recognizes that a result derived from a percentage is an approximation used for real-world estimation, rather than an absolute, exact integer. (p. 1)

1. 理解问题的背景与数学解的表示之间的关系（第 9 项）

分析： 这是本题的核心。学生必须理解表格中“百分比”这一数学表示在现实背景下的含义：它是“电视家庭”与“总家庭”之间的比例关系 (p. 1)。学生需要利用这种理解，将数学计算（除法）与寻找现实世界中实体（瑞士总家庭数）的任务联系起来。

2. 在现实世界背景下解释数学结果（第 1 项）

分析： 题目要求的不是一个抽象的数字，而是对“瑞士家庭总数”的估算。学生完成计算（$2.8 \text{ million} \div 0.858 \approx 3.26 \text{ million}$）后，必须将其回归到现实背景中，从给出的选项中选择最接近的现实估计值（C. 3.3 million） (p. 1)。

3. 识别数学结果或结论在给定问题背景下是否合理（第 2 项）

分析： 在进行选择时，学生需要判断结果的合理性。例如，由于只有 85.8% 的家庭拥有电视，那么总家庭数必然**大于** 2.8 million。通过这种背景常识，学生可以立即排除选项 A，并评估计算出的 3.26 million 是否符合逻辑。

4. 以多种格式解释与情境相关的数学成果（第 5 项）

分析： 题目要求学生处理并转换不同的数据格式：文本描述、表格数据和百分比。学生需要评估表格中的不同列（如区分“电视普及率”和“有线电视订购率”），并根据题目要求（估算总家庭数）选择正确的表示形式进行计算 (p. 1)。

5. 认识到数学概念和数学解的范围与局限性（第 8 项）

分析： 题目要求的是“最接近的估计值”（closest estimate），这反映了对数学解局限性的认识。学生需要明白，基于统计百分比得出的计算结果在现实中往往需要取整或近似，而不是一个绝对精确的数值 (p. 1)。